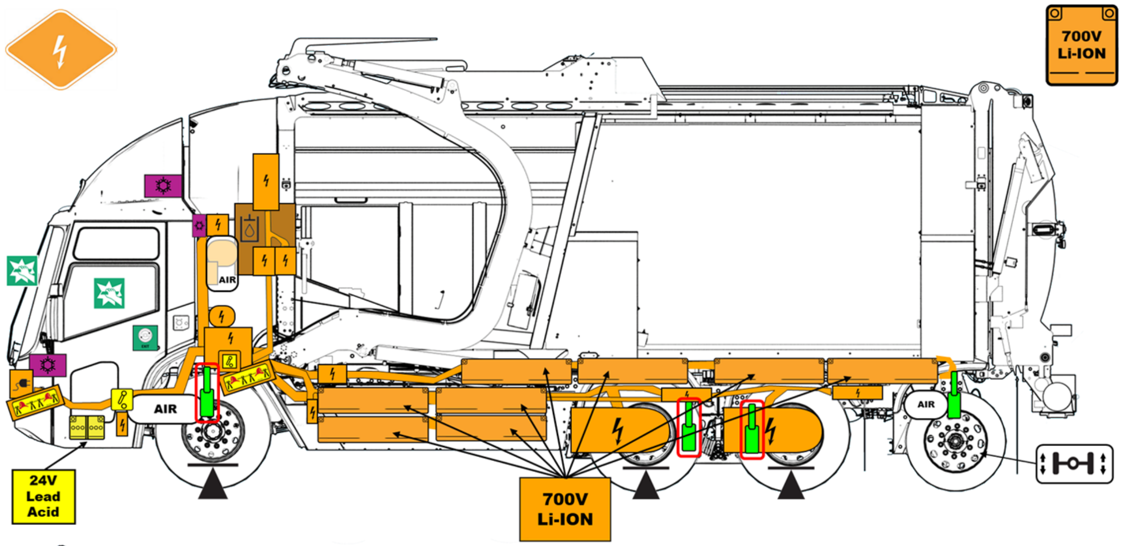
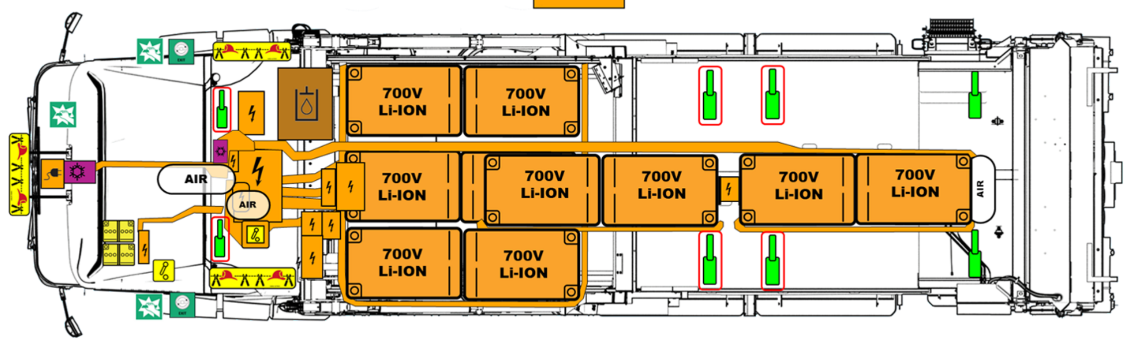
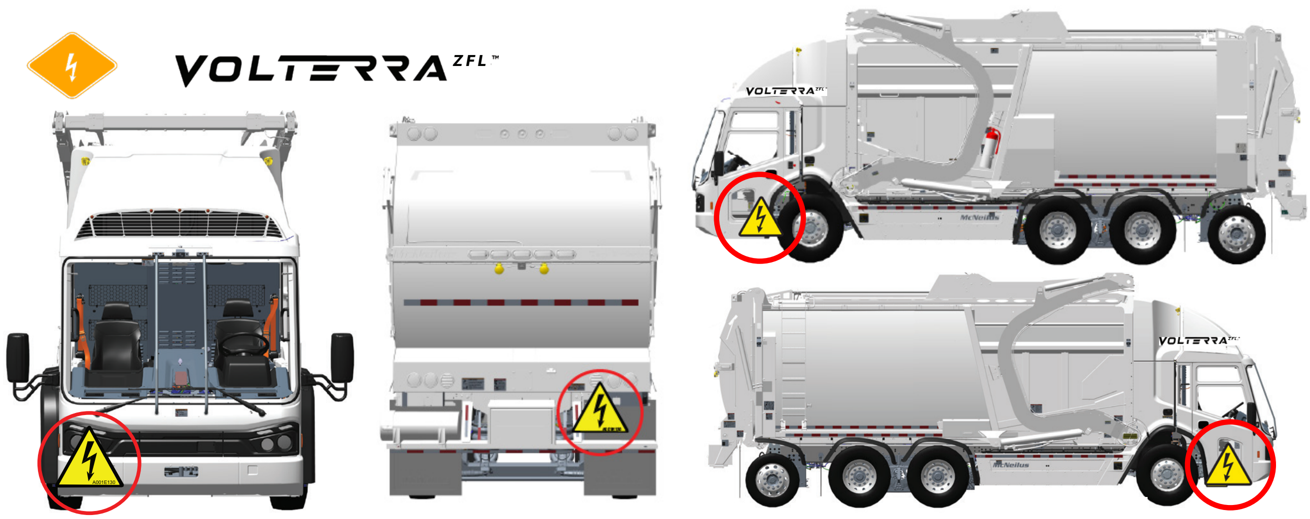
VEHICLE NAME/MODEL: McNeilus® Volterra™ ZFL ™
VEHICLE TYPE/DESIGNATION: Front Loader
PRODUCTION YEARS: 2025 — Present





High Voltage (HV) Battery 620V nominal 700V maximum	Low Voltage (LV) 12V Lead Acid battery - 24V system	High Voltage (HV) Component	Lifting Point	Air Conditioning Component
Low Voltage (LV) Disconnect	AIR Compressed Air Tank	Gas Strut/ Preloaded Spring	High Voltage (HV) Disconnect	EXIT Entry/Exit Points
High Voltage (HV) Power Cable	Moving Tag Axle	Break Points	Hydraulic Fluid Reservoir	High Voltage (HV) Charging Inlet
High Voltage (HV) Disconnect Cut Loop		**NOTE:** These images show information for one variant of this product. The number of traction batteries, location and number of axles, and body configuration can vary depending on the product.		

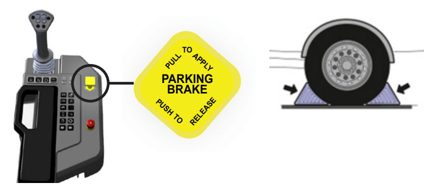
1. Identification / Recognition



2. Immobilization / Stabilization / Lifting

⚠ **WARNING: Lack of engine noise does not mean vehicle is off. Silent movement or instant restart capability exists until vehicle is fully shut down. Wear appropriate PPE.**

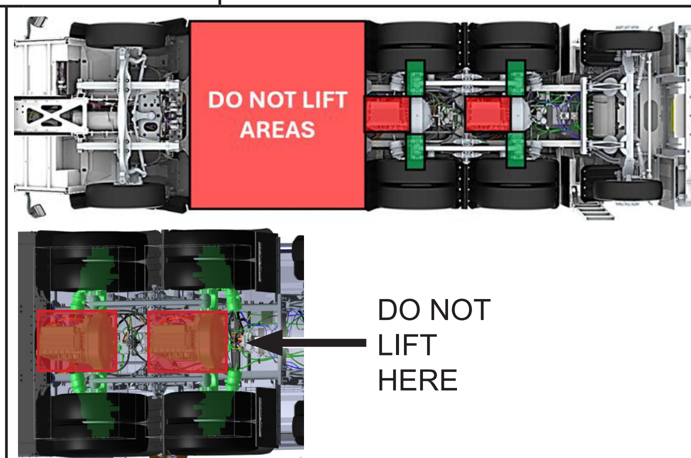
Immobilize the vehicle
Apply park brake (pull to apply).
Chock wheels to prevent vehicle movement. Use only wheel chocks provided with the vehicle located inside the rear toolbox under the tailgate or on either side of the body above the rear fenders.
Use only approved lifting points; avoid marked NO-LIFT zones.



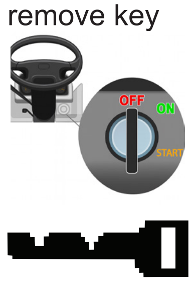
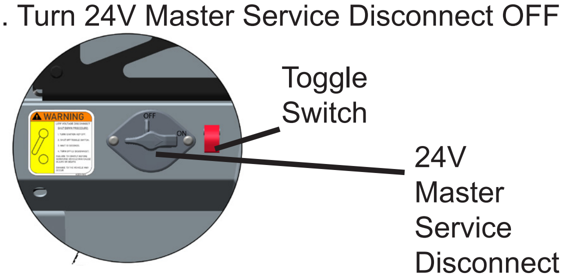
Lifting Points
Use jacking points identified in the illustration. The illustration shows view from underside of the vehicle.

Legend

- Green: Lift/Jack location
- Red: DO NOT LIFT/JACK

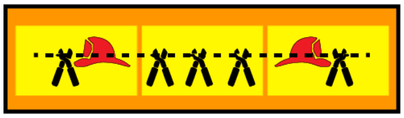


3. Disable direct hazards / Safety regulations

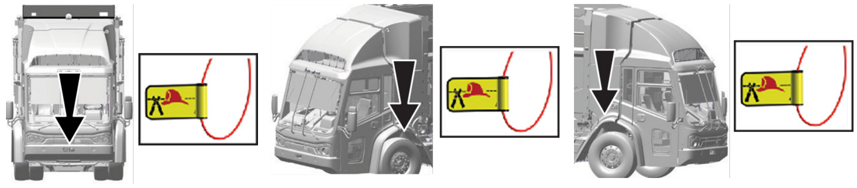
Primary Disabling Method	1. Apply Parking Brake	2. Ignition OFF, remove key	3. Turn 24V toggle OFF, wait 10 seconds 4. Turn 24V Master Service Disconnect OFF
			

Alternative Methods for Disabling the High Voltage System

Option 1: Cable Cut



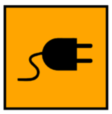
- Locate the most accessible cut loop and cut a section out so the ends cannot touch.



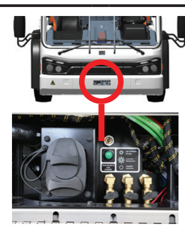
Option 2: High Voltage (HV) Disconnect

- Turn the High Voltage Manual Disconnect switch, located on the street side of the high voltage power distribution unit (HVPDU), counterclockwise to OFF.

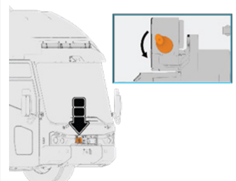
Deactivation Method if Vehicle is Charging



- Press E-STOP on charger (if equipped) or hold green button on vehicle.
- Wait for green light to extinguish, then unplug charger from inlet.
- Turn the 24V Master Service Disconnect Switch to OFF.



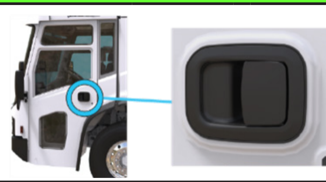
If the inlet is locked:
Stop charging with supply equipment, turn 24V toggle OFF, wait 10 sec. then turn 24V Master Service Disconnect OFF



Mechanically unlock charging inlet if necessary

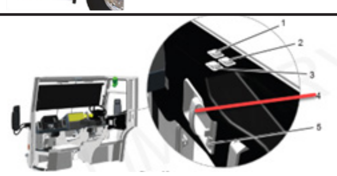
4. Access to the vehicle's occupants

Opening doors from outside.



Pull door handle out toward you and pull door to open.

Opening doors from inside.



Pull door handle toward you and push out on door to open.

Break these windows to obtain access.

Windshield (1) is safety glass per FMVSS and ANSI. Remove Windshield first if accessible. Remove windows (2) as needed.

5. Stored Energy / Liquids / Solids

Component	Type	Hazards Associated
HV battery	700 V LI-ON	
LV battery	24 V Lead Acid	
Refrigerant	R134A	
Hydraulic Oil	ISO 32	
Ethylene Glycol Coolant	50-50	

6. In case of Fire

WARNING: **Fire Hazard. Battery re-ignition is possible, therefore continuous monitoring is critical.**

WARNING: **Inhalation Hazard. Responders should always protect themselves with PPE, including a Self-Contained Breathing Apparatus (SCBA). Furthermore, take appropriate measures to protect civilians downwind from the incident.**

HV battery fires require large volumes of water for cooling and suppression. Standard extinguishers or water streams can be used on the vehicle body, but the battery pack always requires water only.

7. In case of Submersion

WARNING: **Fire Hazard. Flooded battery can still have stored energy and catch on fire. Failure to comply may result in serious personal injury or death.**

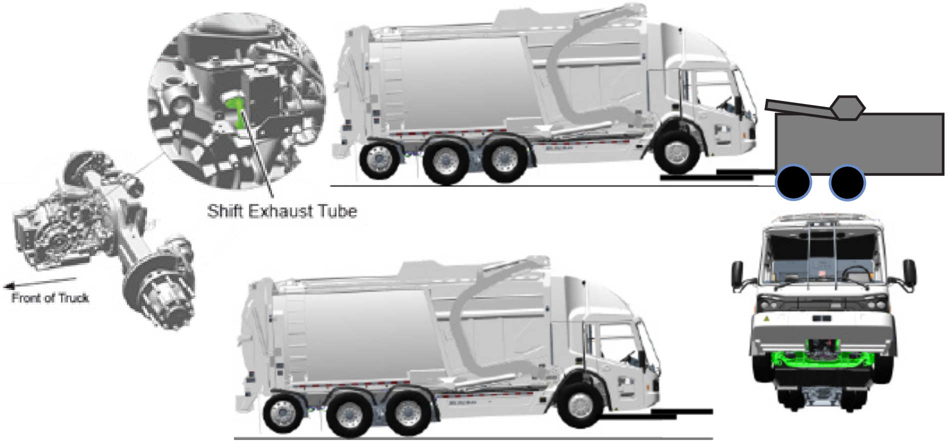
WARNING: **Battery Failure Hazard. Visible bubbling or fizzing from a submerged or damaged electrical vehicle may indicate high-voltage battery failure or an active electrical discharge. Keep your distance until the reaction has stopped. Failure to comply may result in serious personal injury or death.**

A flooded or submerged vehicle does not pose a direct electrical risk to the occupant or surrounding area for extraction.
If possible, remove the vehicle from the water and continue with the disable procedure for this vehicle (Section 3. Disable direct hazards / Safety regulations).

WARNING: **Electric Shock Hazard. Always treat the high voltage components as fully charged and energized. Wear proper PPE when working on vehicle. A flooded vehicle plugged into a charging station can pose an electrocution hazard. Failure to comply may result in serious personal injury or death.**

8. Towing / Transportation / Storage

Store at a safe distance from other vehicles and NOT in an enclosed space. Use front wheel lift for towing. Preferred for both loaded and empty conditions. If selector isn't functional, manually apply air pressure (90-120 PSI) to both rear-axle shift tubes. Confirm center display MAINTENANCE screen states "ALLOWED TO TOW".



CAUTION: Do not tow with wheels on the ground unless axles are disengaged; EV motors may generate electricity.